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EDA

- EDA(Exploratory Data Analysis) is the process of examining,summarization,and visualing data before building machine learning modals.

Why EDA is importannt ¶

Why EDA is importannt :

- Understand the structure of the dataset.
- Identify missing values
- Detect outliers
- Understand feature distributions
- Find relationship between variables
- Check data quality issue
- Prepare data for machine learning

Types of EDA

Types of EDA:

1. Univariate Analysis:
    - Analysing one variable at a time

- Numerical Features Mean, Median, Mode, Standard Deviation, Variance, Histogram,Box plot

- Categorical Features Value counts, Frequency Distribution, Bar chart.

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[3]:

```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("tips")

print(df.head())
print(df.info())
print(df.columns)
```

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
None
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

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```
None
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

[4]:
```
#1. Descriptive Statistics

print(df["total_bill"].describe())
```

```
count    244.000000
mean      19.785943
std        8.902412
min        3.070000
25%       13.347500
50%       17.795000
75%       24.127500
max       50.810000
Name: total_bill, dtype: float64
```

2. Histogram

- A histogram shows the distribution of numerical features
- it divides into intervals(bins) and shows how many observations fall into each interval.

[5]:
```
sns.histplot(df['tip'],bins=20,kde=True)

plt.title("Distributions of Tips")
plt.show()
```

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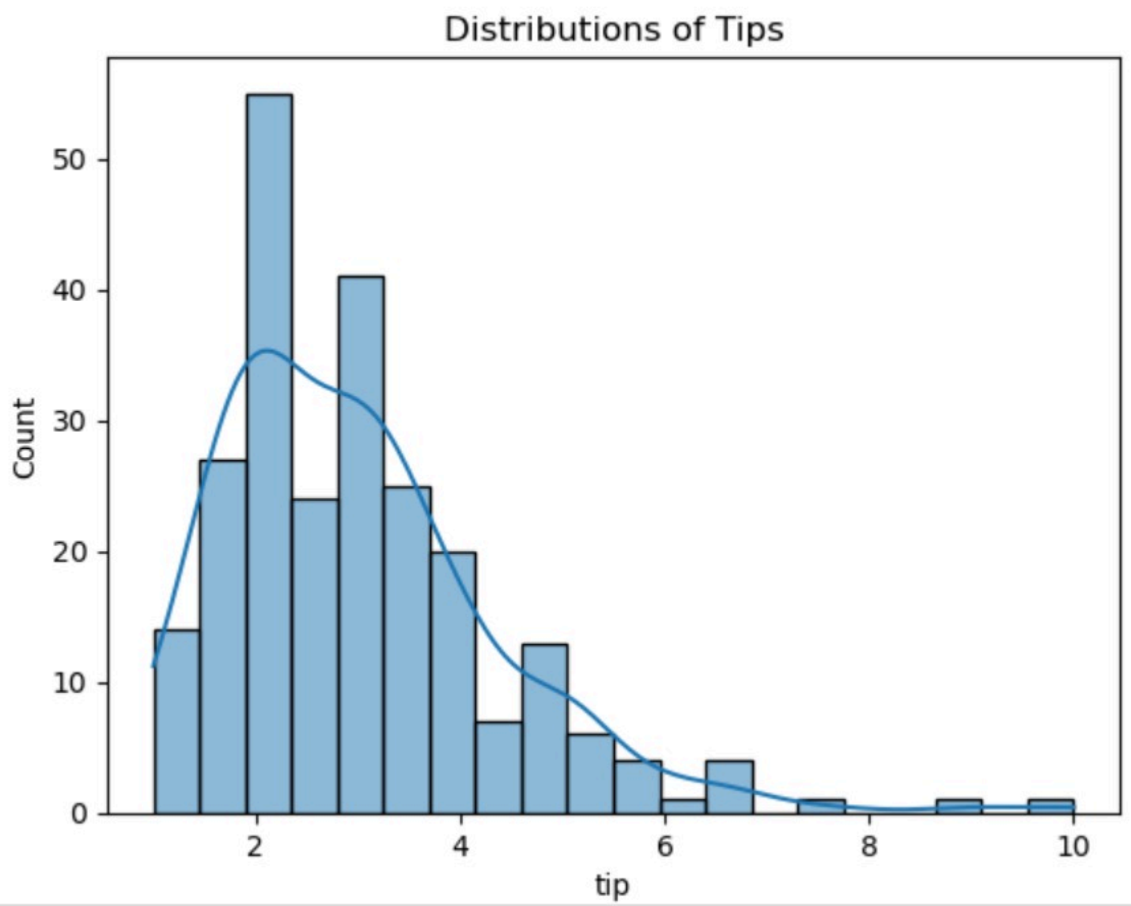
```
plt.title("Distributions of Tips")
plt.show()
```

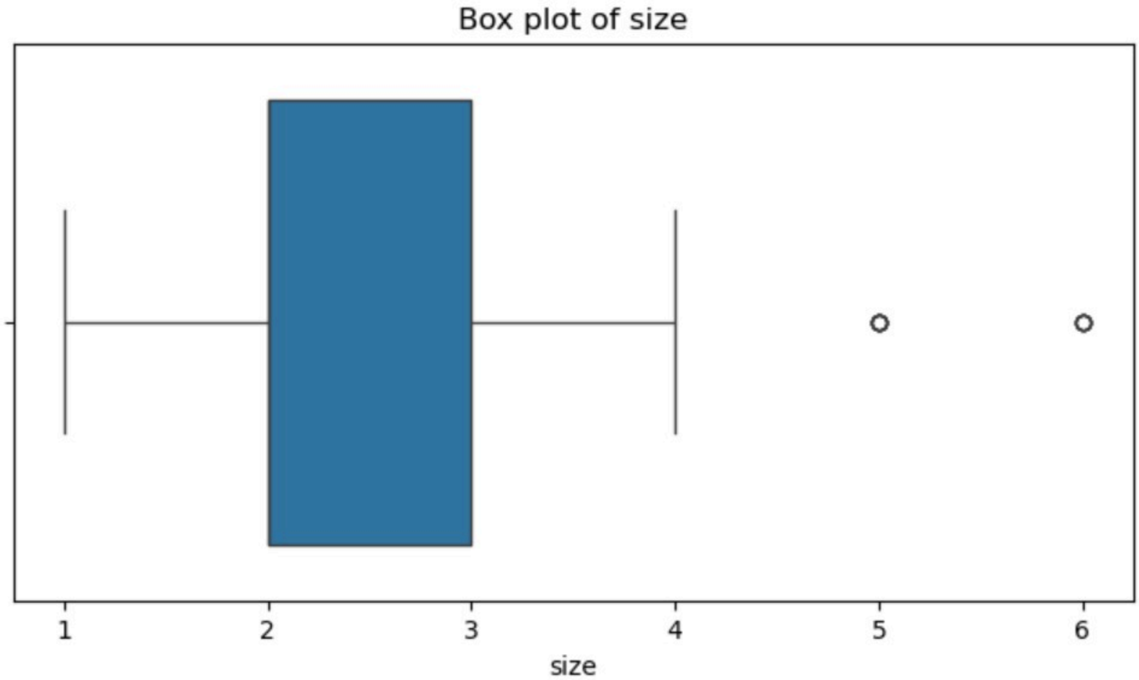


tip

3. Box Plot

[6]:
```
plt.figure(figsize=(8,4))
sns.boxplot(x=df['size'])

plt.title("Box plot of size")
plt.show()
```



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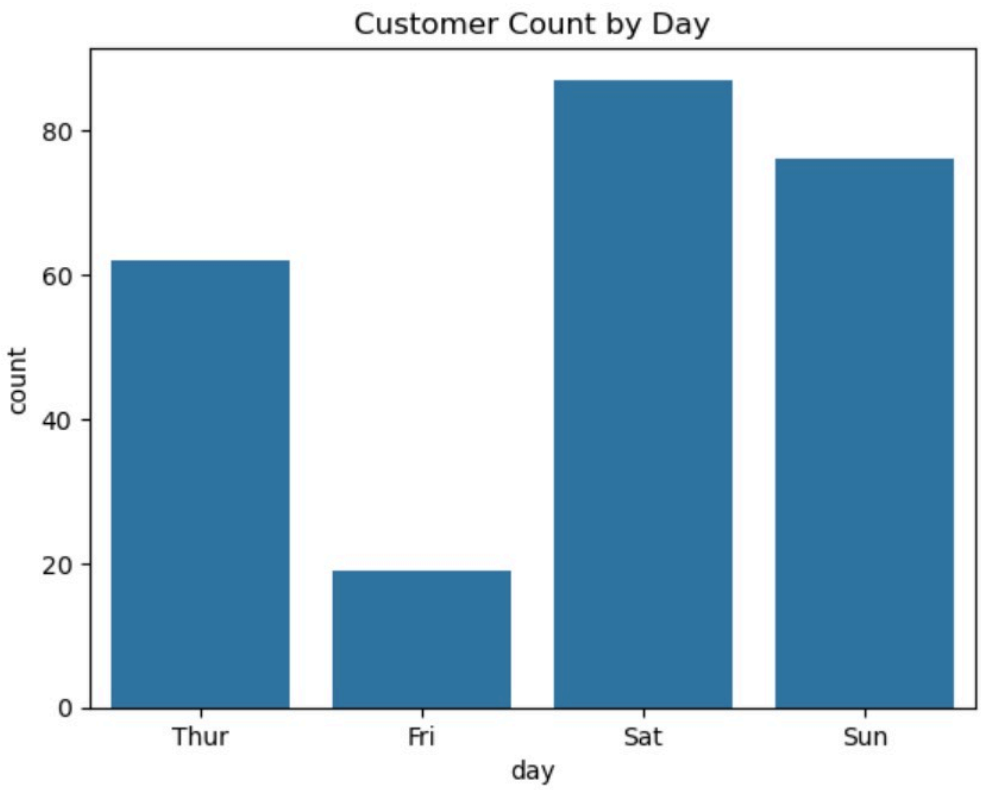
Count Plot

[9]:
```
#Count plot
sns.countplot(x='day',data=df)

plt.title("Customer Count by Day")
plt.show()
```



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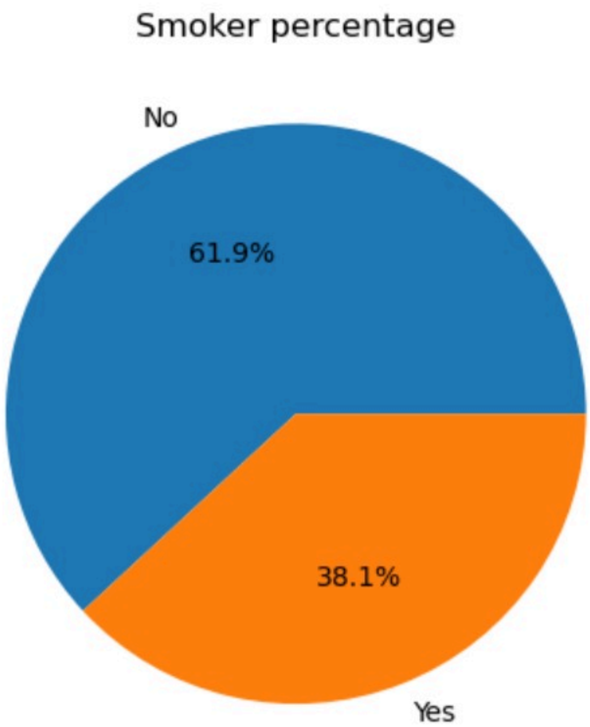
[11]:
```
#Pie Chart  #for yes or now
df['smoker'].value_counts().plot( #yes how many members and no how many numbers
    kind='pie',
    autopct='%1.1f%%'  #for percentage
)

plt.title("Smoker percentage")
plt.ylabel("")
plt.show()
```



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[16]:
```
##Outlier Detection Uisng IQR
Q1=df['total_bill'].quantile(0.25)
Q3=df['total_bill'].quantile(0.75)

IQR=Q3-Q1

lower=Q1-1.5*IQR
upper=Q3+1.5*IQR

outliers_iqr=df[
    (df['total_bill'] < lower) |
    (df['total_bill'] > upper)]

print("IQR Outliers:",len(outliers_iqr))
```

```
IQR Outliers: 9
```

Bivariate Analysis

- Bivariate analysis means applying two variables together to understand the relationship between them.
- Bi=Two
- Variate=Variables

Example:

- Study hours vs Exam Score
- Age vs Salary
- Total Bill vs Fare

Types of Bivariate Analysis

Variale 1 Variable 2 Visualization Numberical Numerical Scatter Plot Categorical Numerical Bar Chart Categorical Categorical Grouped Statistics/Crosstab

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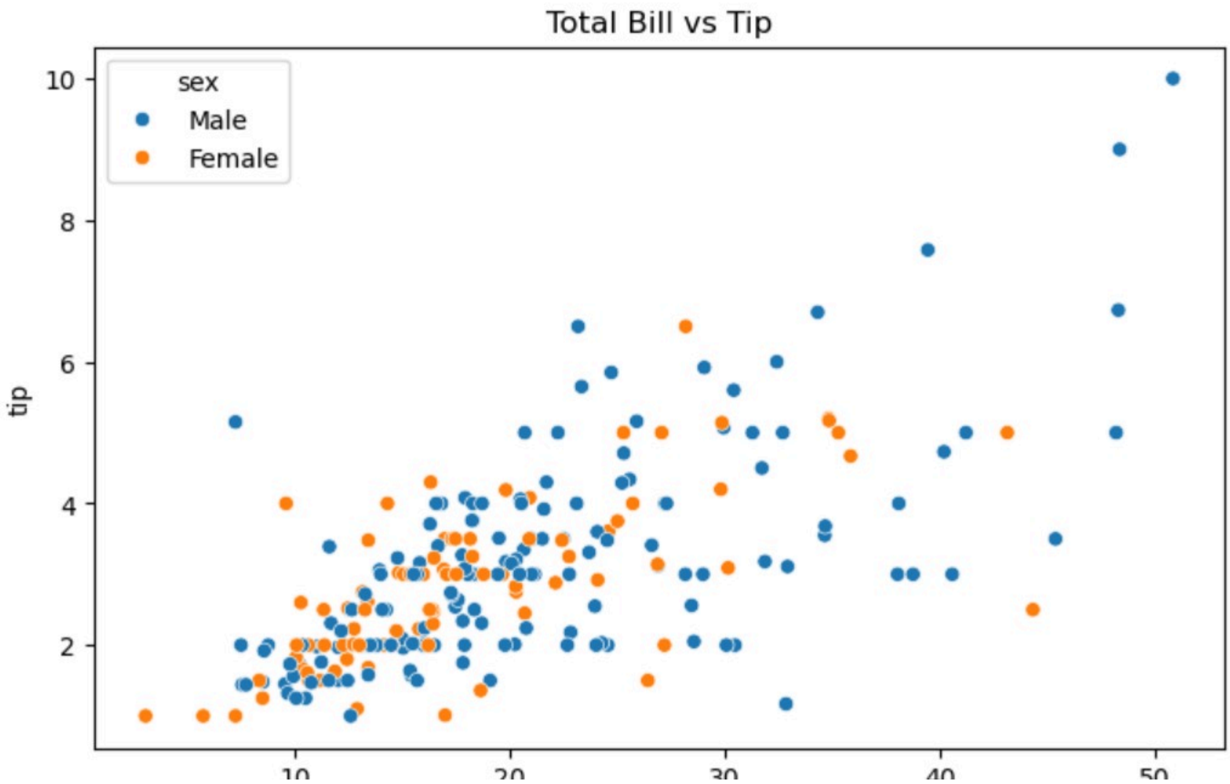
```
[19]: plt.figure(figsize=(8,5))

sns.scatterplot(
    x='total_bill',
    y='tip',
    hue='sex',
    data=df
)
plt.title("Total Bill vs Tip")
plt.show()
```



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Correlation

Pearson Correlation

[23]:
```
corr=df['total_bill'].corr(df['tip'])
print(corr)
```

```
0.6757341092113641
```

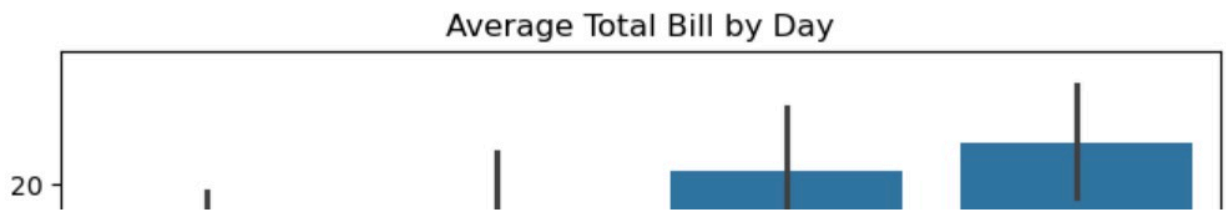
2. Categorical vs Numerical

Day vs Total Bill

- Compare average values accross categories

[25]:
```
plt.figure(figsize=(8,5))

sns.barplot(
    x='day',
    y='total_bill',
    data=df
)
plt.title("Average Total Bill by Day")
plt.show()
```



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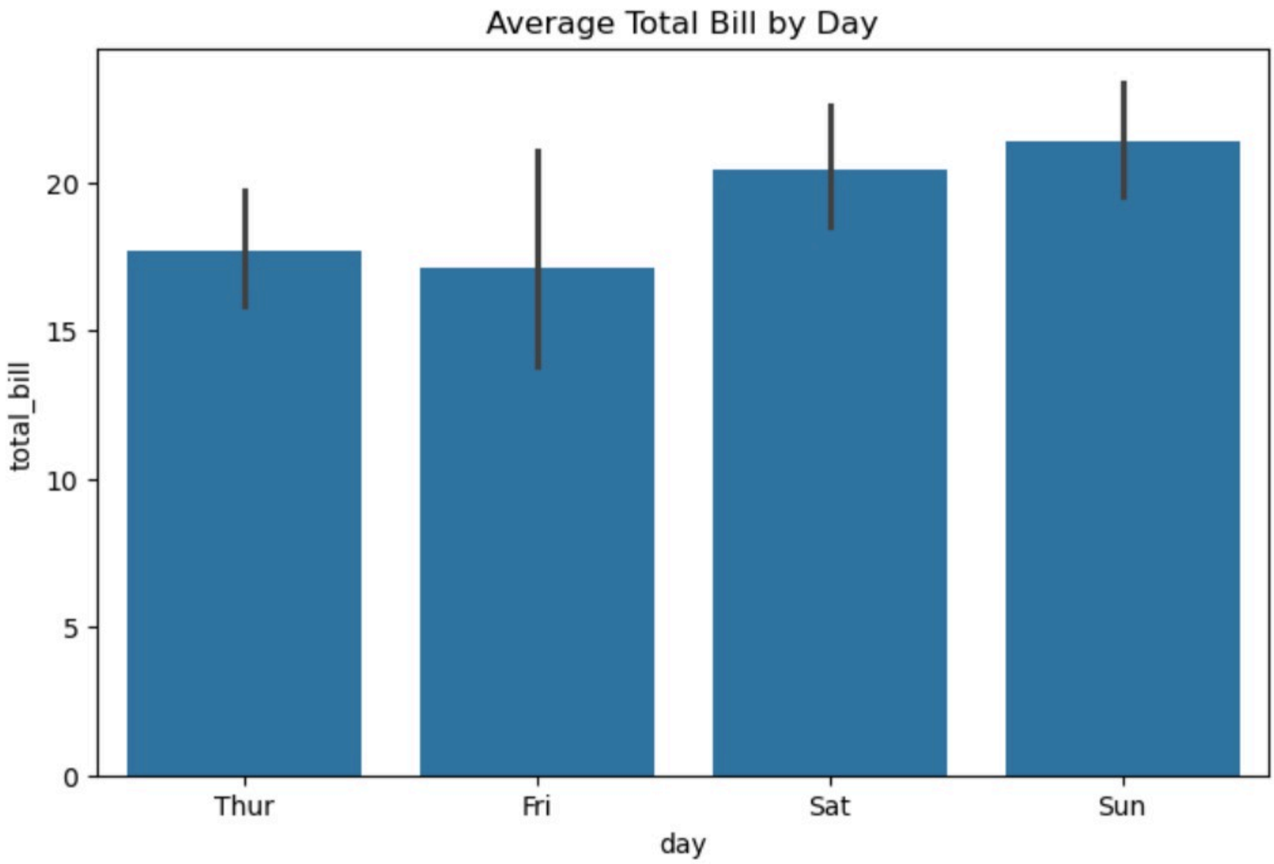
```
)
plt.title("Average Total Bill by Day")
plt.show()
```



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[27]:
```
#Group Statistics
#Mean Bill by Day
print(df.groupby('day',observed=False)['total_bill'].mean())

#Multiple Statistics
print(df.groupby('day',observed=False)['total_bill'].agg(['mean','median','min','max']))
```

```
day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
           mean  median   min    max
day
Thur  17.682742   16.20  7.51  43.11
Fri   17.151579   15.38  5.75  40.17
Sat   20.441379   18.24  3.07  50.81
Sun   21.410000   19.63  7.25  48.17
```

Sex vs Smoker

- Compare frequencies between categories

Crosstab

- Crosstab is used to analyze the relationship between two categorical variables

[30]:
```
#Bar Chart
pd.crosstab(
    df['sex'],
    df['smoker']
).plot(kind='bar')
plt.title("Sex vs Smoker")
plt.xlabel("Sex")
```

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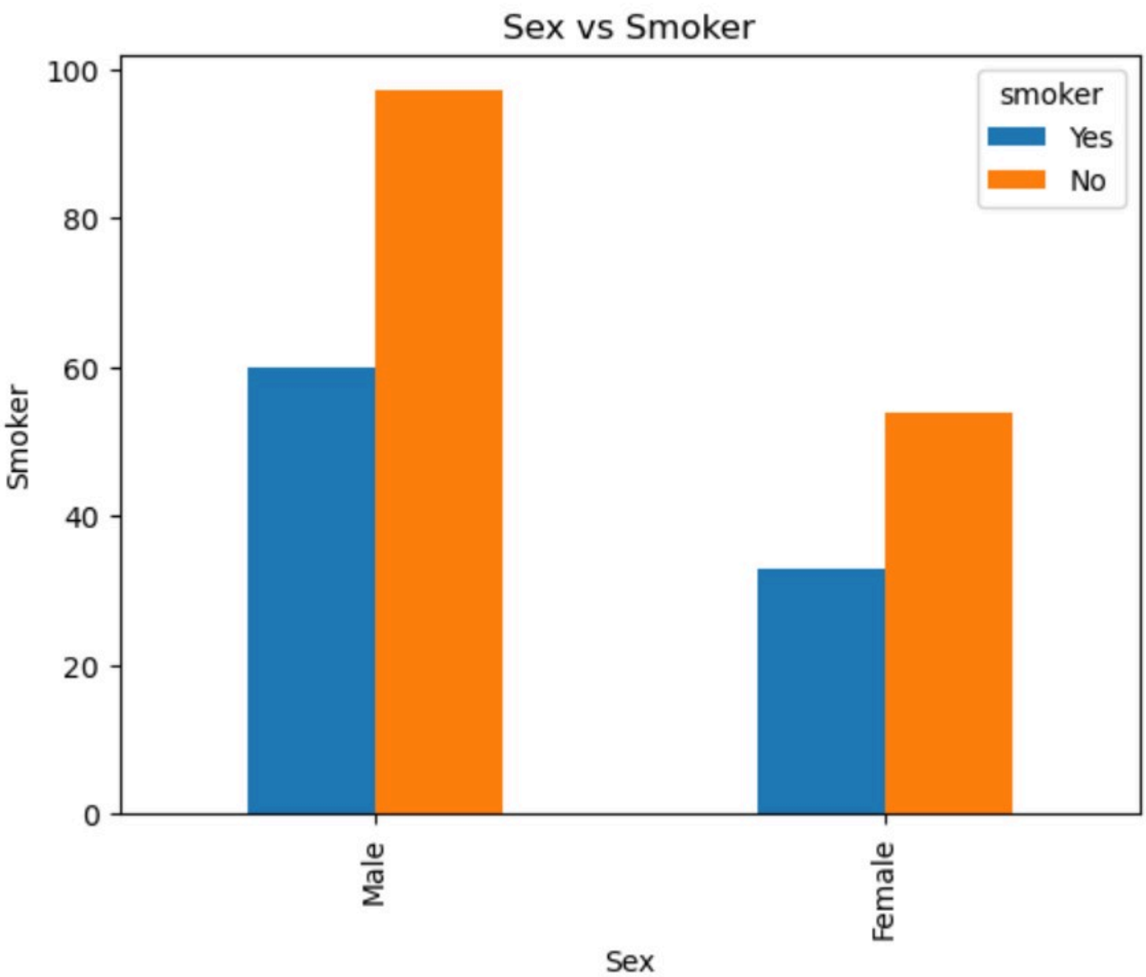
```
).plot(kind='bar')
plt.title("Sex vs Smoker")
plt.xlabel("Sex")
plt.ylabel("Smoker")
plt.show()
```



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Multivariate Analysis

- Multivariate analysis means analysing three or more variales simultanneously to understand the complex relatior data.

Example

Instead of Studying :

- Hight alone(Univariate)
- Hight vs Weight(Bivariate)

We study :

- Height bs Weight vs Age vs Gender
- Ths is multivariate

Objectives of Multivariate Analysis

- Discover hidden relationships.
- Understand interactions among multiple features
- Detect clusters and patterns
- Support feature selection
- Prepare data for Machine Learning

1. Pairplot

- A pairplot compares every numerical feature with each other numerical feature.

- It automatically creates

    1. Scatter Plot

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It automatically creates

1. Scatter Plot
2. Histogram
3. Relationship matrix

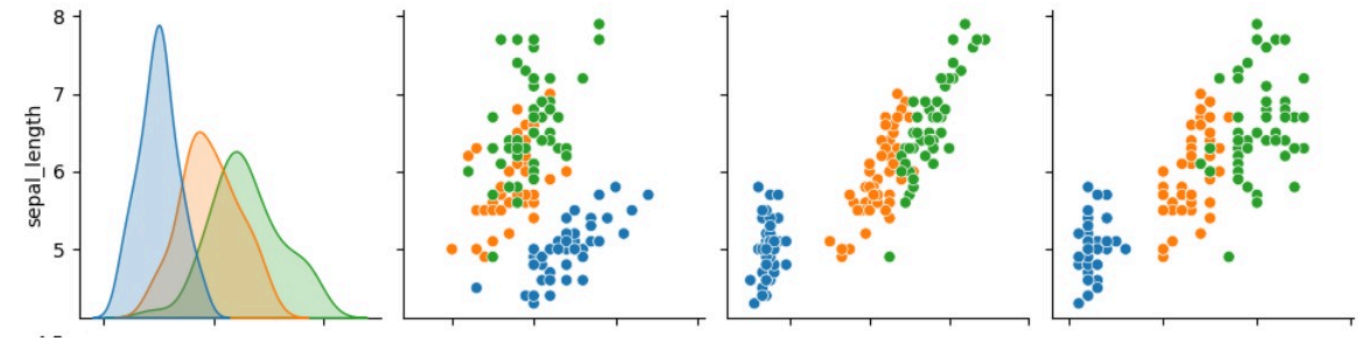
Why Use Pairplot?

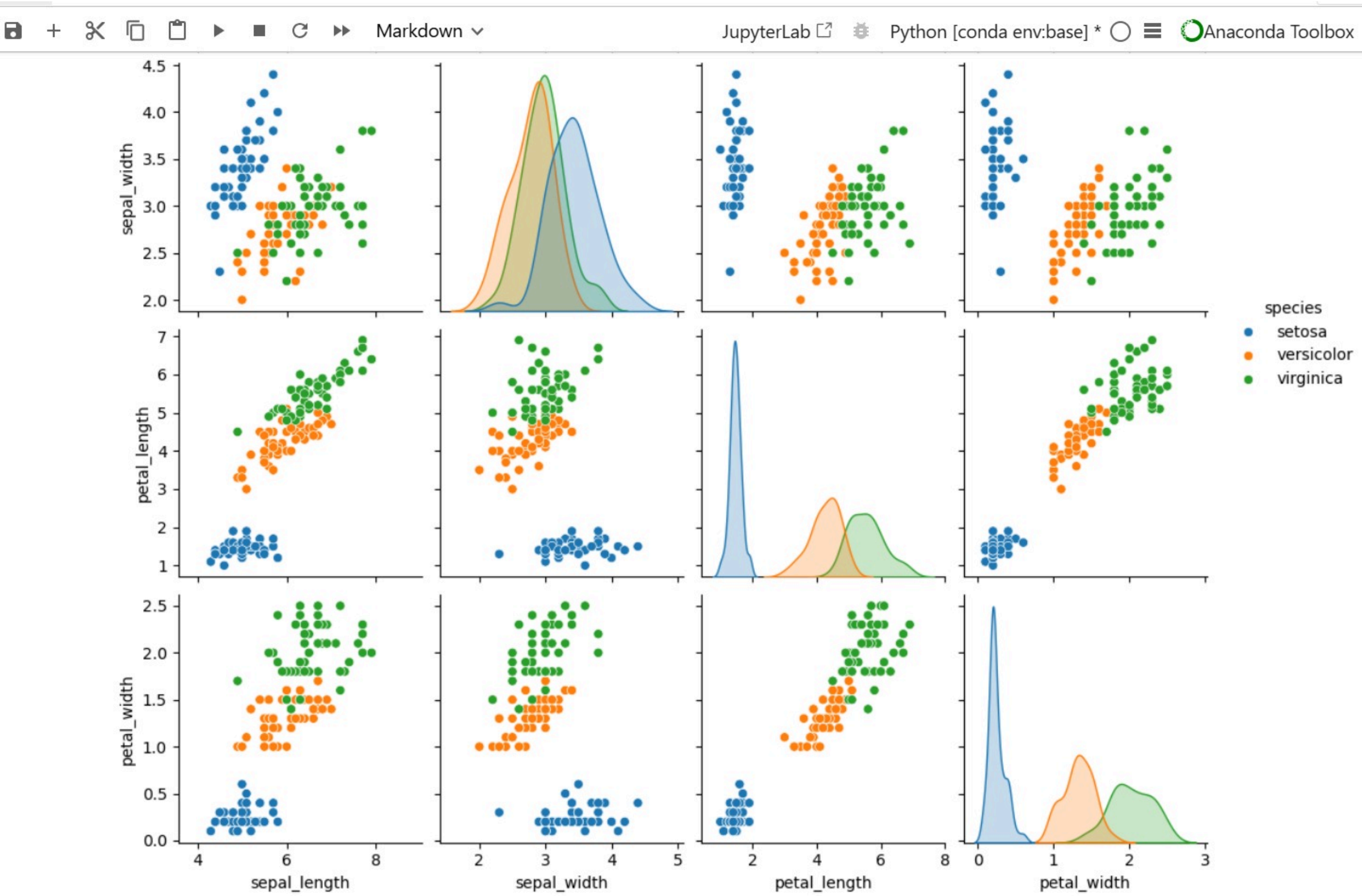
1. Detect correlations
2. Find Clusters.

[34]:
```
#pairplot
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("iris")

sns.pairplot(
    df,
    hue='species'
)
plt.show()
```

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sepal_width
petal_length
petal_width
sepal_length
sepal_width
petal_length
petal_width
species
setosa
versicolor
virginica

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2. Color-Coded Scatter PLOT(3 rd Dimensions)

- A scatter plot normally shows:
    1. X-axis - variable1
    2. Y-axis - variable2

A third variable can be represented using colors:

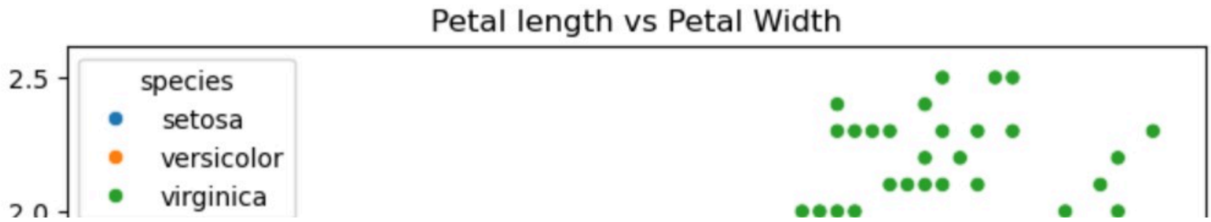
Example:

```
1. X- Petal Length
2. Y- Petal Width
3. Color- Species
```

[37]:

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x='petal_length',
    y='petal_width',
    hue='species',
    data=df
)
plt.title("Petal length vs Petal Width")
plt.show()
```



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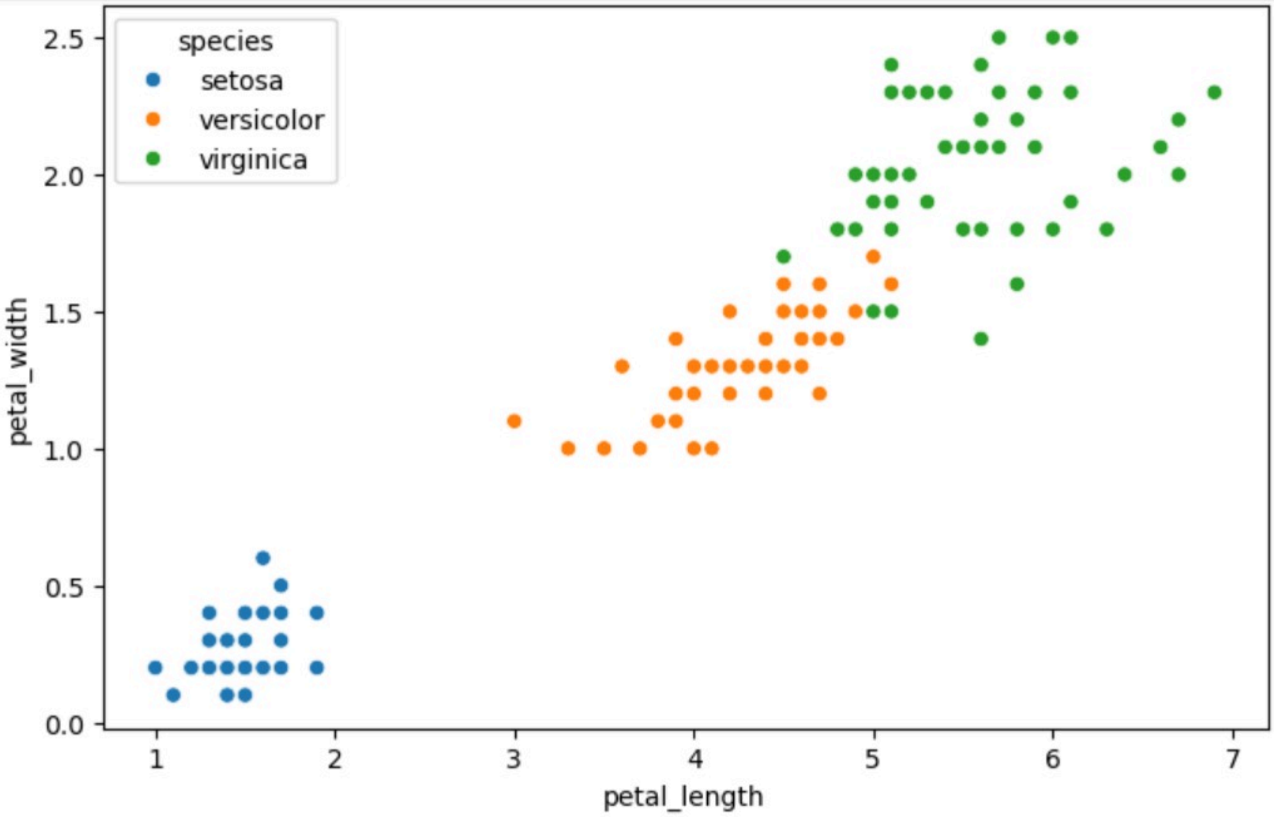

3. Scatter Plot

- A 3D Scatter Plot uses: 1.X-axis 2.Y-axis 3.3D axis

[38]:

```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt

fig=plt.figure(figsize=(8,6))
```

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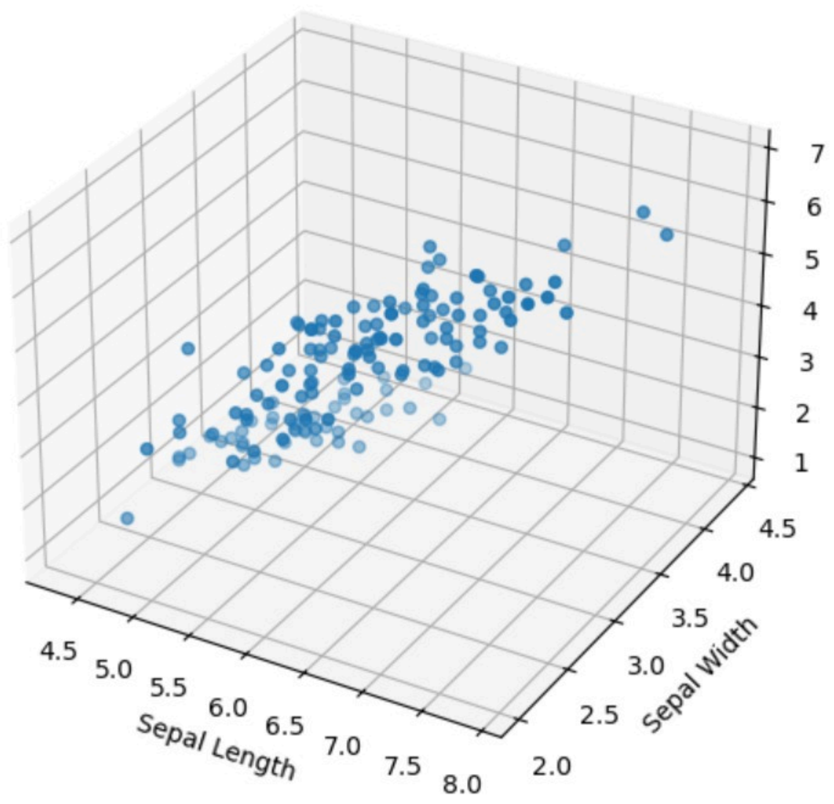
```
    df['petal_length']
)
ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
ax.set_zlabel('Petal Length')

plt.show()
```



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Correlation Analysis

What is Correlation ?

- Correlation measures the strength and direction of relationship between two numerical variables.

Correlation range

- -1<=r<=1

[41]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df=sns.load_dataset("iris")

print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

Correlation matrix

- A Correlation matrix shows the correlation etween all numerical columns.

[43]:

```
corr_matrix=df.corr(numeric_only=True)
print(corr_matrix)
```

```
              sepal_length  sepal_width  petal_length  petal_width
sepal_length      1.000000    -0.117570      0.871754     0.817941
```

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sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

Heatmap

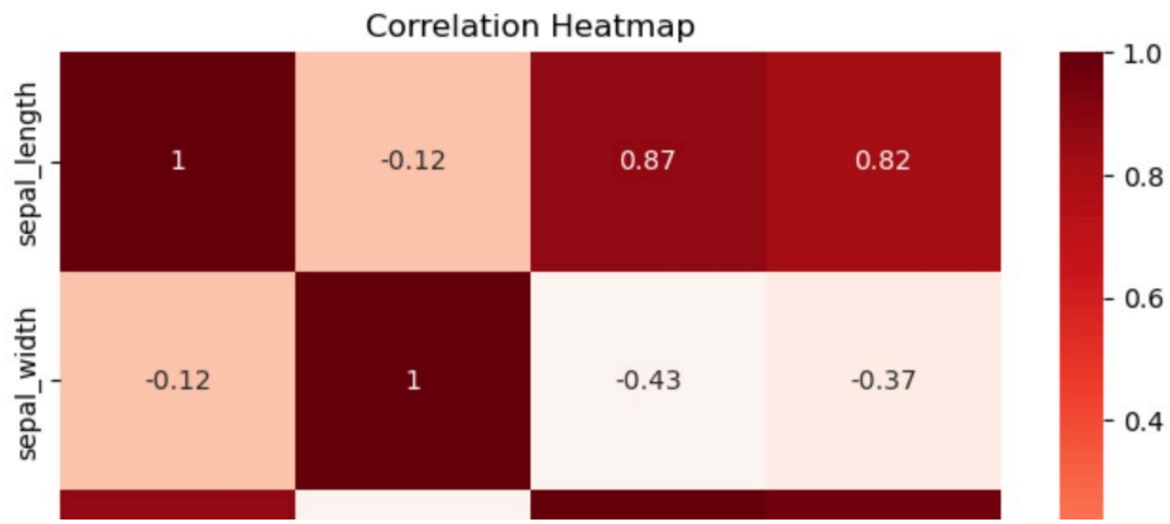
- A heatmap visualises

[52]:

```
plt.figure(figsize=(8,6))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Reds'
)

plt.title("Correlation Heatmap")
plt.show()
```



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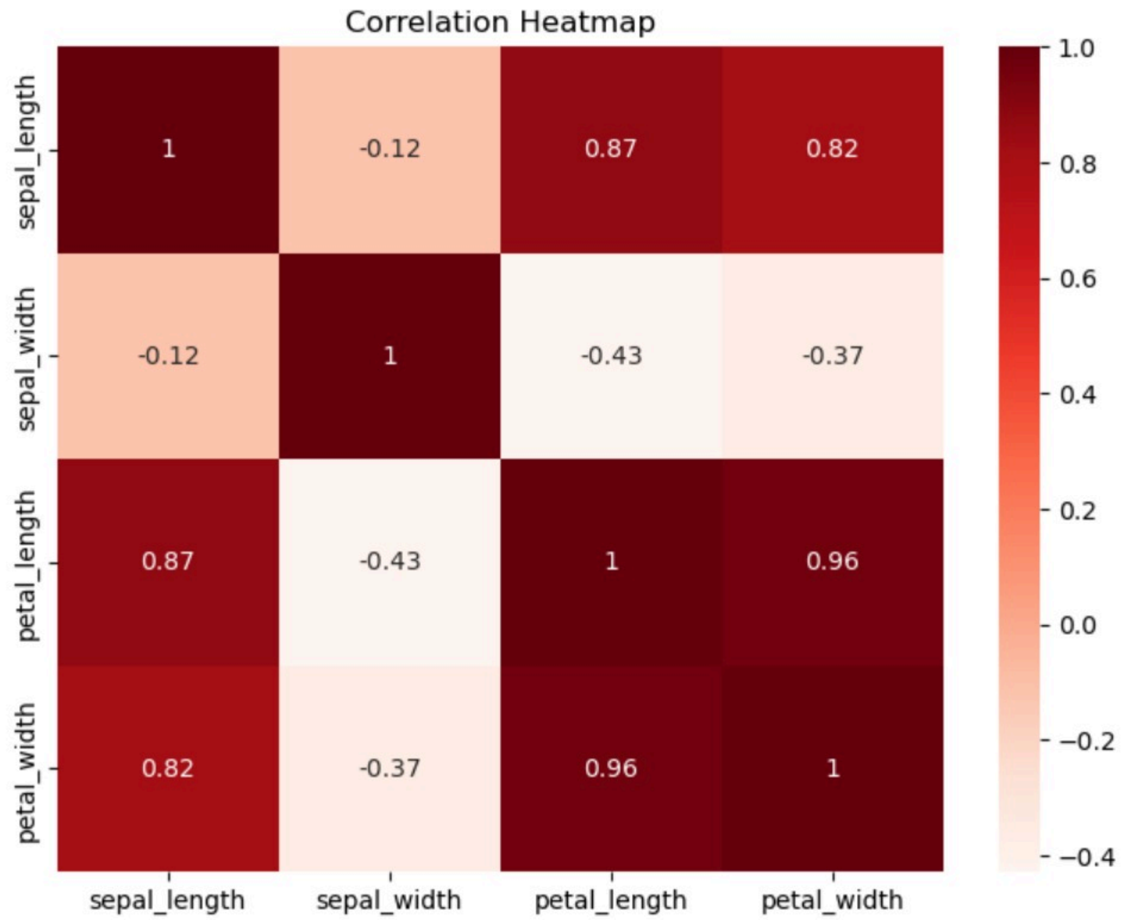
```python
plt.title("Correlation Heatmap")
plt.show()
```



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VIF(Variance Inflation Factor) for MultiCollinearity Detection

What is Multicollinearity ?

- Multicollinearity occurs when indepedent variables(features) are highly correlated with each other.

Example

```
* Suppose a house price dataset contains :
1. House Area(sq.ft)
2. Number of Rooms
3. Number of Bedrooms

* These features may contain similar information and be highly correlated
* This creates multicollinearity
```

What is VIF? ¶

- Variance inflation Factor(VIF) measures how much the variance of a featire is inflated because of its correlation with the other feature.

VIF Interpretation Table

1. VIFValue 1. Interpretation
2. 1 2. No Correlation
3. 1-5 3. Low Correlation
4. 5-10 4. Moderate Multicollinearity

5.

> 10 5. Severe Multicollinearity

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[55]:
```python
from statsmodels.stats.outliers_influence import variance_inflation_factor

X=df.drop("species",axis=1)

vif=pd.DataFrame()

vif["Feature"] = X.columns

vif["VIF"]=[
    variance_inflation_factor(X.values,i)
    for i in range(X.shape[1])
]
print(vif)
```

```
        Feature         VIF
0  sepal_length  262.969348
1   sepal_width   96.353292
2  petal_length  172.960962
3   petal_width   55.502060
```

[61]:
```python
X_new=X.drop("sepal_length",axis=1)

vif_after=pd.DataFrame()

vif_after["Feature"] = X_new.columns

vif_after["VIF"]=[
    variance_inflation_factor(X_new.values,i)
    for i in range(X_new.shape[1])
]
print(vif_after)
```

```
        Feature        VIF
0   sepal_width   5.856965
1  petal_length  62.071308
```

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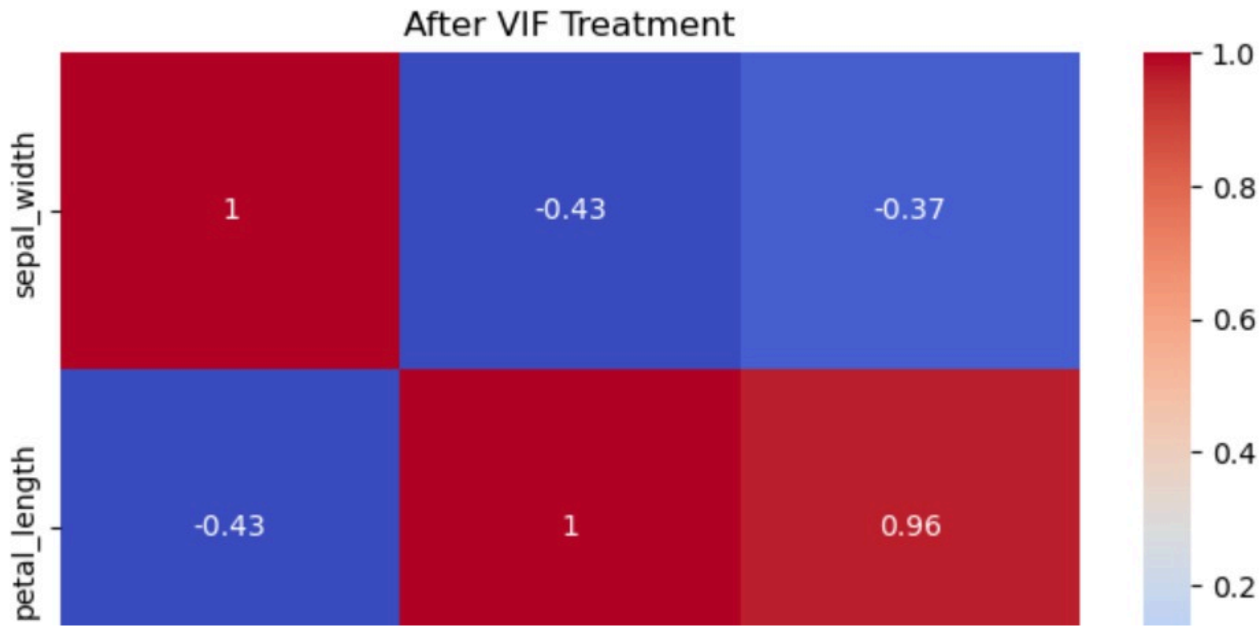
```
1  petal_length  62.071308
2   petal_width  43.292574
```

[63]:

```python
plt.figure(figsize=(8,6))

sns.heatmap(
    X_new.corr(),
    annot=True,
    cmap='coolwarm'
)

plt.title("After VIF Treatment")
plt.show()
```



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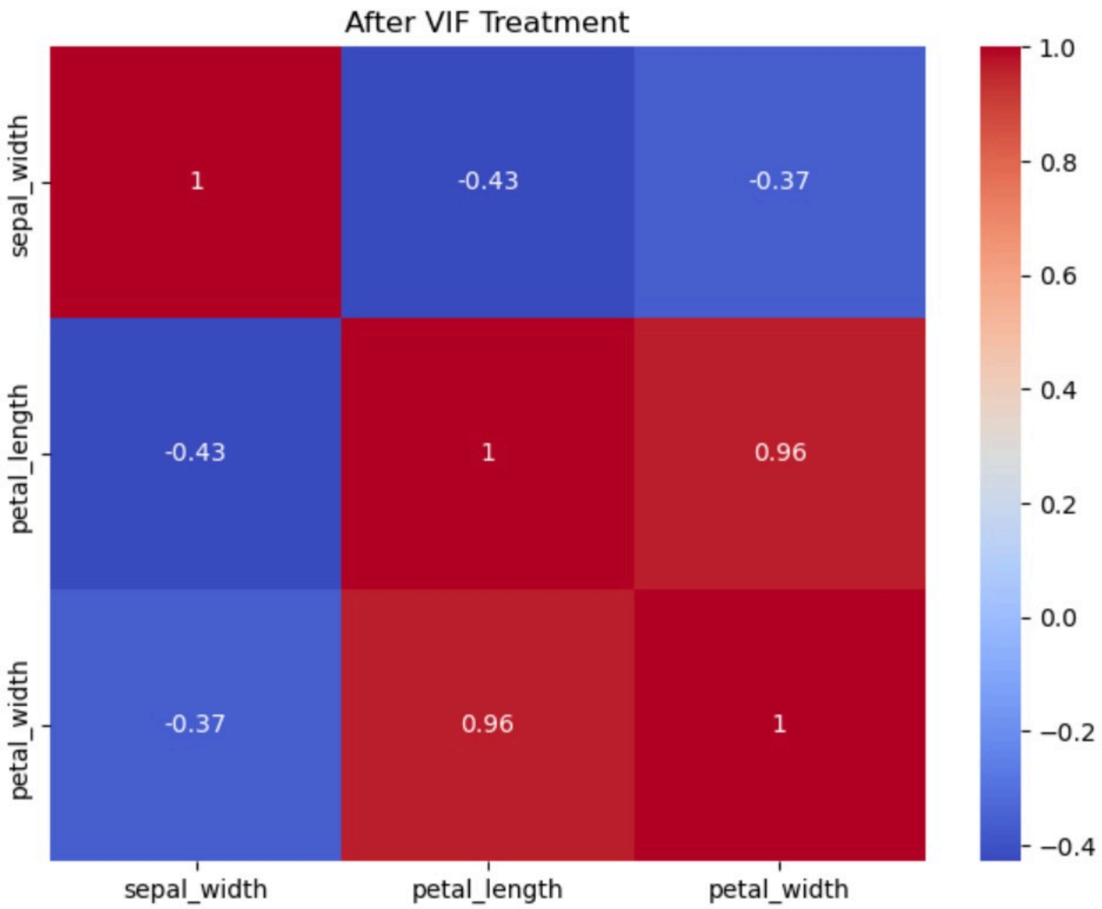
```
plt.show()
```



[ ]: